

Ashmit Dutta

612-323-1292

✉ ashmitd2@illinois.edu

🌐 ashmitdutta.xyz

📘 [ashmitdutta](#)

🔗 [aedutta](#)

EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

Bachelor of Science, Computer Engineering

Aug 2023 – May 2027

- Accepted paper at ICML 2025 workshop on [finetuning Chain of Thought in LLMs](#).
- Built a 2-way superscalar out-of-order CPU ([ECE 411](#)) in SystemVerilog; **top 10** in the class design competition.

EXPERIENCE

Nablon AI

Jun 2026 – Present

AI Engineering Intern

Remote — San Francisco, CA

- Building **Sentinel**, an automated LLM-evaluation framework (FastAPI + async Python workers on Azure) for an enterprise automotive client; designed a 5-tier scanner pipeline flagging jailbreaks, PII, and toxicity.

Assured and Trustworthy AI Research Lab – University of Illinois

Dec 2025 – Present

ML Research Intern (Prof. Huan Zhang)

Urbana, IL

- Implemented the **Softmax** operator in [auto-LiRPA](#), an open-source neural-network verification library (VNN-COMP winner); wrote C++ patches **cutting peak GPU memory 44%** and Python sanity-checking pipelines across 16+ benchmarks (ResNets, ViTs, YOLO, control nets).

John Deere

Jan 2025 – Jan 2026

Software Engineer Intern

Champaign, IL

- Shipped a React + Spring Boot data-exporting platform **to production** for **500+ internal users**.
- Built core REST APIs and CI/CD pipelines over PostgreSQL and DB2 handling **millions of records**.
- Designed a multithreaded job scheduler **cutting DB search time 3x**; added AWS CloudWatch observability.

University of Chicago Data Science Institute — DSI Summer Lab

Jun 2024 – Aug 2024

ML Research Intern (Fermilab collaboration)

Chicago, IL

- Built a reusable multi-modal [Graph Neural Network library](#) (PyTorch Geometric) for Fermilab's Exa.TrkX neutrino reconstruction, training over **20,000+ detector events** on a Slurm HPC cluster.

PROJECTS

Embedded Drone Controller (Skylink v1.0) | STM32, C, bare-metal

Jan 2026 – May 2026

- Wrote bare-metal firmware for an STM32L0 (ARM Cortex-M0+) handheld controller on a self-designed 4-layer PCB.
- Sampled dual analog joysticks via on-chip ADC and a Bosch BMI270 IMU over **I²C**; streamed 8-byte control packets over a sub-GHz TI CC1101 **SPI** transceiver.
- Reworked the RF matching network from 433 to 315 MHz and verified the radio link end-to-end on hardware.

Unix-like RISC-V Operating System | C, RISC-V, QEMU

Mar 2025 – May 2025

- Built a Unix-like OS from scratch with a **trap-based preemptive scheduler**, Sv39 virtual memory, and UART/VIRTIO/RTC device drivers.
- Implemented syscalls, ELF program loading, and a read/write filesystem, debugged with GDB/QEMU.

Autonomous Drone Racing | ROS, Python, C++, PyTorch

Oct 2025 – Dec 2025

- Built a ROS 2 control stack (min-snap planner + geometric controller) for a Crazyflie 2.1 drone.
- Tuned controller gains via Bayesian optimizer; **outperformed 3 published baselines** on real hardware.
- Designed a voltage-feedback compensator to hold thrust under battery sag; debugged tracking error via flight logs.

LEADERSHIP, PUBLICATIONS & AWARDS

Co-Founder & Lead Organizer, [Online Physics Olympiad \(OPhO\)](#): founded the largest student-run international physics competition (20k+ community); led website development and a team of 20+ students.

Awards & Leadership:

- Winner (Modal Inference + Bright Data AI tracks), [Stanford TreeHacks 2026](#) — out of 380 teams.
- 2nd Place, Bradley Mottier Innovation Challenge, UIUC ISE (\$2,000) — [AR glasses for ASL translation](#).
- Executive Board Member (Treasurer), [IEEE Chapter at UIUC](#).
- Undergraduate Teaching Assistant, CS 450 (Numerical Analysis) — Publications: [Google Scholar](#)

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript, SQL, SystemVerilog, RISC-V Assembly

Embedded & Systems: 32-bit ARM Cortex-M microcontrollers / STM32, bare-metal firmware, preemptive scheduling, I²C/SPI/UART, ADC, closed-loop / real-time control, PCB design, RTL/FPGA, GDB/QEMU

Testing & Verification: HDL test benches, unit testing, Spike/RVFI co-simulation, hardware bring-up, CI/CD

Software & Infra: Python, PyTorch, CUDA, REST APIs, PostgreSQL, Docker, Git, CI/CD, AWS, Linux